Beyond Believability: Evidence-Linked Context Representations for Behaviourally Grounded Social Agents

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Abstract

Social agents can sound human without reconstructing the humans they represent. We test whether an evidence-linked relational-behaviour graph, alone or combined with memory and reflection, improves grounded reconstruction over a compact persona or memory alone. From approximately 230 minutes of edited couples-therapy material, we build representations from earlier episodes and evaluate 400 condition-blind predictions for 5 participants on later held-out situations. Three architectures lead for different participants in this small sample; the hybrid never ranks first, and mechanistic fidelity diverges from observable fidelity. Graphs support provenance and explicit conditional pathways, but errors increase when meaning is carried by surrounding partners or non-verbal behaviour omitted from transcripts. These preliminary findings point toward evidence-calibrated combinations of memory, structured mechanisms and multimodal evidence for behaviourally grounded, psychologically plausible agents. The relevant question is not which persona is most convincing, but which evidence and representation are adequate for the inference being made.

1 Introduction

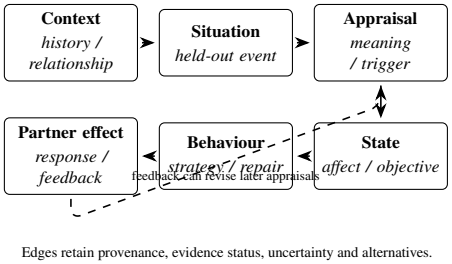
Believability is a common success criterion for social agents, but it does not establish fidelity to the target. A fluent response may be reasonable in general yet unsupported by the target's history, relationships or observed actions [16, 18, 8]. This gap matters when agents stand in for real people or are aggregated into simulated populations: individually plausible errors can change interactions and downstream conclusions.

We ask a simple empirical question: *what kind of context helps an LLM reconstruct a person's later behaviour?* We compare a compact persona, chronological memory with reflection, an evidence-linked behavioural graph, and a memory–graph hybrid. Rather than asking whether an output merely "sounds right", we hide a later observed response and score whether the model recovers its interpretation, affect, objective, action, escalation or repair, and immediate effect on a partner.

The distinction is between *psychological plausibility*—a defensible account of why someone might respond—and *behavioural grounding*—evidence that this person responded in this way. Achieving behavioural grounding depends on the evidence available to the agent and how that evidence is represented. A prediction may fail because the representation obscures an important mechanism, or because the source material never captured the decisive signal. Adding more text cannot recover omitted gaze, posture, prosody, silence, timing or relationships; yet providing too little evidence can invite generic assumptions.

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a Evidence-linked relational pathway



b Mean fidelity (/12): no universal winner

	A	B	C	D	E	Pooled
Minimal	7.55	6.85	**6.75**	4.10	**6.25**	**6.30**
Memory	**8.65**	6.85	5.00	2.70	5.15	5.67
Graph	7.75	**7.25**	5.70	**4.40**	5.70	6.16
Hybrid	8.00	6.80	5.30	2.30	6.00	5.68

Figure 1: Evidence-linked pathway (a) and mean fidelity by condition (b). Bold cells mark column maxima; pooled values are descriptive.

We contribute (i) a provenance-aware relational-behaviour graph—a structured map connecting a person's history and situations to interpretations, affect, relational objectives, behaviour and partner effects, with each connection tied to its source and uncertainty—(ii) a temporal holdout protocol separating mechanistic from observable fidelity, and (iii) an empirical comparison that, within this small sample, complicates our initial expectation that the hybrid would be strongest. Different representations lead for different participants, and richer context sometimes supports a persuasive but wrong story. The result reframes context design as an evidence-selection problem rather than a contest to build the most elaborate persona.

This proof-of-concept methodology proposal establishes these three tools and surfaces questions—architecture heterogeneity, the mechanistic–observable gap and missing modalities—for larger multimodal evaluation. Its participant-level results suggest that meaningful capture of human behaviour is conditional on the person and available evidence, not a fixed property of an architecture.

2 Related work

Behavioural science motivates longitudinal representation: past behaviour can inform future behaviour through repeated routines and intentions, while personality research locates consistency in situation-dependent "if–then" signatures rather than static traits [11, 10]. Generative Agents established memory and reflection as a basis for believable agents [12]; subsequent work grounds agents in interviews, behavioural histories and psychological patterns [13, 6, 17]. Persona and graph-based approaches add structure [9, 14, 7], but do not generally retain the provenance, evidential status, competing explanations and revisions behind each behavioural link. Our graph treats those features as part of the representation, making situation–mechanism–behaviour pathways inspectable. We test whether that auditability also improves held-out reconstruction, alone or with memory, while prior evaluations show weak prompt-only alignment with observed behaviour [4, 16, 8].

3 Evidence-linked context representations

Source and construction. The source comprises nine approximately 26-minute episodes (about 230 minutes) from Season 4 of *Couples Therapy*, following nine participants across four relationship systems [5]. This is a useful stress test because the same people return across time, describe their own motives, respond to partners, receive competing interpretations and sometimes revise earlier accounts. It therefore supports both longitudinal evidence construction and a genuine later holdout, while making relational errors visible: predicting one person often requires understanding what another person did or meant. AI-assisted transcription and preliminary structuring were followed by human speaker correction and longitudinal cross-checking. We retain individual accounts alongside relationship-level overlap and disagreement, adapting dyadic Framework analysis [3]. The programme is edited and performative, not diagnostic truth.

A behavioural pathway connects context, situation, appraisal, internal state, behaviour and partner effect (Figure 1a). Missing components remain missing and competing interpretations may coexist.

Evidence status distinguishes direct self-report, observed behaviour, partner report, clinical formulation, accepted formulation, recurrence and researcher inference. Demographic membership never

Table 1: Held-out case: a plausible prediction adds confrontation absent from the observed repair-oriented response.

Situation shown to the agent	Observed response (sealed)	One blinded prediction
Participant E's partner could invite E to an event with another anchor partner, but takes no action. The missed invitation is discussed in therapy.	E notices that something is blocking the partner, asks how it can be resolved, and avoids adding anger because anger would not help.	E might say: "You had the choice to include me and nothing happened. I feel forgotten and not genuinely considered. I need to understand why, and I need a clearer plan."

creates a behavioural edge without explicit source support. Immutable episode graphs preserve what was knowable at each point; a unified graph accumulates corroboration, contradiction and revision. This relational/systemic framing treats one partner's behaviour as another's situation and feedback [15, 2].

Controlled representations. All conditions derive from one pre-holdout evidence set: (i) **minimal persona**: sparse role, relationship and explicitly stated goal information; (ii) **memory-reflection**: chronological observations, embedding-based retrieval, importance scoring and reflection cycles following Park et al. [12]; (iii) **behavioural graph**: evidence-linked conditional pathways without the full chronology; and (iv) **hybrid**: memory/reflections plus graph pathways. All use gpt-5.6-luna with medium reasoning effort, identical system instructions, situation, schema and 700-token output limit. Memory retrieval uses text-embedding-ada-002. Input lengths differ because representational form is the present ablation; matched-budget comparisons remain future work.

4 Held-out evaluation

Episodes 1–7 provide development evidence. For each participant, four interpersonal situations from Episodes 8–9 are frozen before the participant's interpretation or response; the remaining situation is presented identically to each condition. The 5-participant analysis contains 20 unique participant–situation cases across one dyad and all three differently positioned members of a triad. Four conditions generate five repetitions per situation, yielding 80 predictions per participant and 400 total. Repetitions estimate output variability, not independent human observations.

Each prediction contains situation interpretation, affective stance, behavioural strategy, relational objective, escalation/repair and immediate partner effect. Each dimension is scored 0 (mismatch), 1 (partial) or 2 (match), producing a 0–12 total. Interpretation, affect and objective form a *mechanistic* subscore (0–6); strategy, escalation and partner effect form an *observable* subscore (0–6). Outputs are shuffled and condition labels concealed. One human annotator reviews all dimensions, starting from rubric-based proposed scores and documenting amendments. Ratings are frozen and checksummed before the condition key is opened.

5 Results

In this sample, no architecture consistently dominates (Figure 1b). Memory-reflection leads for A, the graph for B and D, and the minimal persona for C and E; the hybrid never leads. Pooled means are 6.30 for minimal persona, 6.16 for graph, 5.68 for hybrid and 5.67 for memory-reflection. These five-participant estimates admit sampling variability; we report no significance tests and treat the pattern as hypothesis-generating.

Representation effects differ by dimension. Minimal persona has the highest pooled mechanistic (3.24/6) and observable score (3.06/6), but only narrowly exceeds graph (3.20 and 2.96). For D, minimal persona is strongest mechanistically while graph is strongest observably and overall; for E, hybrid is marginally strongest mechanistically while minimal persona is strongest observably. Thus a representation can generate a psychologically plausible interpretation yet predict the wrong action, escalation or partner effect.

Situation effects are also pronounced. In E's possible-relocation case, observable fidelity is only 0.8–1.2/6 across architectures despite stronger mechanistic scores: predictions often reconstruct surrounding partners incorrectly, while the transcript contains limited non-verbal evidence. C likewise communicates uncertainty through visual and temporal cues incompletely preserved in text. These cases expose a shared modality limitation rather than a failure unique to the graph.

6 Discussion

Three findings make the negative result informative. First, apparent representation fit varies across participants, although five people cannot distinguish a person-by-architecture interaction from noise. A's meanings are layered autobiographically; B's situation-to-response explanations are comparatively explicit. Minimal persona leads for C and E, while the graph leads for B and D; this does not establish that either representation is generally better. When decisive evidence is missing, compact representations have fewer opportunities to elaborate the wrong mechanism. The hybrid concatenates memory and graph without learned fusion or matched budgets, so its result may reflect longer, unweighted context rather than a general limit of combination.

Second, plausible mechanisms and observable behaviour come apart. The minimal persona leads both pooled subscores narrowly, yet participant-level rankings diverge: for D, it best captures mechanism while the graph best captures observable behaviour. In the held-out example (Table 1), the prediction recognises exclusion and a need for clarity, but turns E's measured attempt to understand the partner into a more confrontational demand. A system evaluated only for plausibility could count this as success; the observed strategy, escalation level and likely partner effect say otherwise.

Third, the target is not the whole unit of prediction. C's hardest case involved an external partner outside the focal triad; E's errors often came from assumptions about surrounding partners. C's gaze, pauses, laughter, bodily orientation and shifts of attention were also weakly preserved in text. Richer text could therefore become more specific without becoming better informed; combining memory and graph did not cross this evidence boundary.

These observations suggest a selective context stack: memory for autobiographical explanation, graphs for conditional pathways and relational position, and audiovisual models for gaze, prosody, timing or posture. Selection must be evidence-calibrated; indiscriminate addition may multiply unsupported inferences. The broader research claim is diagnostic: reconstruction depends on alignment among human expression, captured evidence, represented relationships and the prediction task. Although drawn from five individuals rather than a demographic sample, the same plausible-versus-grounded calibration applies to population-level fidelity; the mechanistic/observable split offers one test before scaling to synthetic populations.

7 Limitations, ethics and conclusion

Broadcast therapy is mediated evidence, not complete lived or clinical truth. Transcripts omit or discretise facial expression, gaze, gesture, touch, prosody, silence and interaction timing; annotations such as "long pause" cannot preserve their continuous interpersonal distribution. All conditions therefore share an information ceiling, and combining two text-derived representations cannot recover a missing modality. Video may add evidence but remains shaped by camera selection and editing.

The sample is purposively small because longitudinal graph construction favours depth per participant over cast breadth. One annotator provides no inter-rater reliability estimate, and proposed scores may anchor adjudication despite blinding. Input budgets are unmatched; the base model may contain programme information despite neutral identifiers and contamination controls. Therapist formulations are interpretations, and the graph adopts a relational/systemic lens. Public availability does not remove privacy, dignity or dual-use risks [1]; the method is not diagnostic or validated for consequential decisions. We excluded cast members whose material centred on severe trauma to avoid modelling people through their most vulnerable disclosures.

This proof of concept shows that the graph, temporal holdout and mechanistic/observable split expose distinctions worth testing rather than settling architecture choice. Larger multimodal studies should test whether missing-cue errors can be recovered and whether architecture heterogeneity persists. The question is which evidence and representation support the behavioural inference.

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A Reproducibility materials

Code, representations, protocols and experimental artefacts are available at `https://github.com/rokiseydi/beyond-believability`. The repository contains the evidence-status codebook; canonical pre-holdout evidence tables; the four representation packages with token counts; prompts, model identifiers, seeds and generation parameters; all blinded outputs; frozen human-adjudicated ratings and checksums; blinding keys; scoring rubric; additional graph figures; and episode provenance. Blinding keys are provided for audit but were not opened until ratings were frozen.

B Evidence status and scope

Edges are labelled as direct self-report, observed behaviour, partner report, clinical formulation, accepted formulation, repeated observation, literature-supported hypothesis or researcher inference. Alternative pathways are retained where evidence conflicts. No edge is created from demographic membership alone. Future work should use consented multimodal longitudinal data, participant review, multiple annotators, matched-budget ablations and explicit governance for access to high-fidelity behavioural agents.